A Fully-Local, Open-Source ASR Pipeline for Bengali Climate-Survey Interviews

Two-Witness Fusion + LLM Reconciliation

MMI Bangladesh

June 22, 2026

Fully local · Open-source · Privacy-safe · Reproducible

The Problem

The Challenge

Goal. Turn coastal-Bangladesh survey interviews into clean, searchable transcripts *with English translation*.

- **Hard audio.** Field recordings transcoded to 8 kHz, $\sim$12 kbps (telephony-grade); heavy compression.
- **Colloquial, regional Bengali.** Dialect forms, fast speech, code-switching with English banking / survey terms.
- **Faithfulness is paramount.** Numbers, dates and entities must never be invented — a survey cannot turn **250** into **2.5**.
- **Data privacy.** Sensitive interviews $\Rightarrow$ the system must run **fully local** and **open-source**.

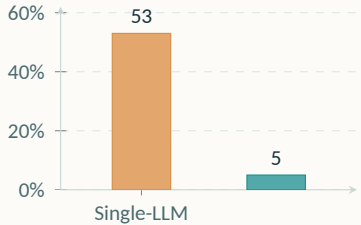
Hard constraint

Everything runs on a **single laptop GPU (8 GB VRAM)**. No cloud. No API.

Why a New Pipeline?

What does not work

- A *single* ASR engine makes too many errors on this audio.
- A *single* LLM "clean this up" prompt hallucinates — rewriting text that was already correct.
- Dedicated NMT (IndicTrans2) corrupted numbers on our data ($6 \rightarrow 696$, $250 \rightarrow 35$).



Character error rate — naive LLM rewrite

vs. a constrained pipeline (*literature, illustrative*).

Our design — two complementary witnesses reconciled by an LLM judge

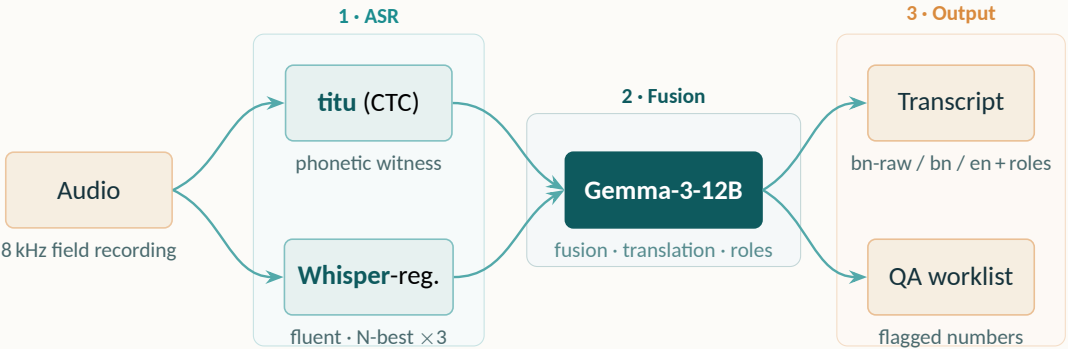
Two witnesses. A phonetic engine + a fluent engine (ROVER / generative error-correction).

LLM judge. Reconciles them into one faithful transcript *and* translates — one constrained call.

N-best. Multiple readings give evidence to disambiguate (*ward* vs. *world*).

The System

The Pipeline



Models load *sequentially* (load → use → unload) to fit a single 8 GB GPU.

Output flows raw ASR → corrected Bengali → English, with **ENUMERATOR** / **RESPONDENT** roles.

The Models

Role	Model (Hugging Face)	Type
Witness 1 phonetic	`hishab/titu_stt_bn_fastconformer`	NeMo FastConformer **+ CTC**
Witness 2 fluent	`bengaliAI/tugstugi_bengaliai-regional-asr_whisper-medium`	Whisper-medium **Bengali regional fine-tune**
Judge fusion	`gemma-3-12b-it` Unsloth Q4_K_XL GGUF	instruction-tuned LLM

All three run fully local — no cloud, no API. Privacy by construction.

Two Witnesses, Two Failure Modes

titu — phonetic (CTC)

Transcribes *acoustically*, token-by-token, with little language-model bias.

+ **Faithful to sound** — reliable for the *sound* of names, numbers, uncommon dialect words.
− Can run words together or look garbled.

Whisper — fluent (seq2seq)

Strong internal language model in the decoder.

+ **Fluent, well-structured** output.
− LM bias swaps a content word for a plausible-but-wrong one: বন্যা “flood” → বোন না “sister-no”.

Why pair them

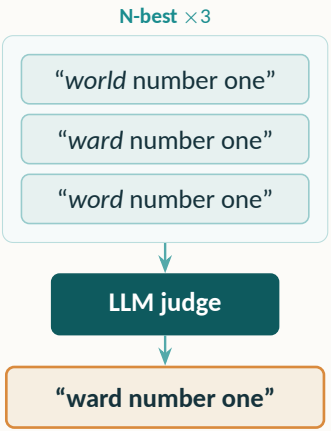
titu anchors the **sound**; Whisper anchors the **structure**. Their errors are largely **independent** — where one fails, the other often holds, so fusion recovers more than either alone (ROVER / complementary-systems). The *regional* fine-tune matters: on coastal dialect, a dialect-tuned *medium* model beats even a generic large-v3.

Why N-best, Not Top-1

A single **top-1** from each engine commits to one reading. On an ambiguous word — *ward* vs *world*, or a number — top-1 can be wrong with **no recourse**.

N-best gives the judge several candidate readings, so it can pick the right one using **context** + the **phonetic anchor**.

This is **Generative Error Correction** (GER / HyPoradise): feeding an LLM the N-best beats correcting a single transcript. We use **seeded sampling** for genuine diversity — plain beam search collapsed to one reading.



The judge picks the contextually correct reading.

An LLM Judge, Not a Vote

Classic ROVER does **word-level voting** — mechanical and context-blind. An LLM judge does what voting cannot:

1. **Reasons with context** — “ward” not “world”; recognizes a bank name; reads a dialect form by meaning.
2. **Reconciles** the two witnesses into one coherent Bengali line — not a Frankenstein of votes.
3. **Translates** to faithful English *and* **assigns roles** — all in **one constrained call**, fully local.
4. **Resists hallucination** via tight rules (keep numbers exact, code-switch only on genuine English) — where a naïve “clean this up” prompt rewrites correct text (**53%** vs **5%** CER).

In one line

Two ASR engines that fail *differently* give independent evidence; N-best widens that evidence; and an **LLM judge** can actually *reason* over it — reconcile, translate, label — instead of blindly voting.

Key Design Decisions

Decision	Why
Gemma-3-12B (Q4)	Best Bengali model that fits 8 GB; beats a 4B model on faithful numbers (**250**, not **2.5**).
Two witnesses	Phonetic (**titu**) anchors *sound*; fluent (**Whisper**) anchors *structure* — their errors cancel.
N-best fusion	Multiple Whisper readings let the judge pick the right word (*ward* vs. *world*).
Content-based roles	Speaker labels from *what is said* (Q vs. A), not error-prone voice clustering.
Constrained prompt	Explicit rules: translate meaning, keep numbers exact, code-switch only on genuine English.

Fully local · fully open-source · deterministic and reproducible.

Evidence & Roadmap

Results — Coherence



Fully coherent — 28/50: clean prose (e.g. “so many people’s conditions worsen during floods”).

Partial — 21/50: gist clear, rough spans — a mistranslation (“murug fish” for chickens), a transliteration, or an `[unclear]`.

Garbled — 1/50: largely incoherent.

Honest assessment

Only **56%** is clean prose; another **42%** is usable but needs a careful read. Numbers are **flagged separately** — this measures *coherence*, not numeric accuracy. Single LLM rater; a Bengali-speaking re-rate would make it human-validated.

Built-in Quality Assurance

The pipeline emits a **review worklist** — only the segments a human should check:

- any segment containing a **number**;
- any segment where the two witnesses **disagree**;
- any text the model marked **[unclear]**.

Reviewers verify **a handful of flagged spots** — not all 200+ segments. Targeted, efficient QA.

Why it matters

The audio-quality ceiling cannot be fully removed locally. Instead of hiding it, the pipeline **surfaces** exactly what to double-check — honest, auditable output.

Next Steps — To Production

1. **Compute upgrade (16 GB GPU)** — the single biggest lever:
   - no memory paging $\Rightarrow$ fusion at full speed ($\sim$1 hr $\rightarrow$ $\sim$15 min per interview via batching);
   - run **Gemma-3-27B** (does not fit 8 GB) $\Rightarrow$ better number-faithfulness.
2. **Batch processing** — run all interviews unattended.
3. **Human-in-the-loop QA** — analysts verify flagged numbers against audio; the worklist drives this.
4. **Optional frontier benchmark** — a private Claude / Opus track to spot-check the open-source pipeline's accuracy.

Status

The pipeline is **complete and validated**. Remaining work is *hardware* (16 GB GPU) and *operations* (batch + QA) — not research.

Summary

- A **fully-local, open-source** Bengali ASR + translation pipeline, privacy-safe by design.
- **Two-witness fusion + LLM reconciliation** delivers faithful, readable transcripts with English translation and speaker roles.
- **Honest, auditable accuracy:** strong for qualitative analysis; numbers auto-flagged for targeted human QA.
- **Production = compute + operations**, not new research — a 16 GB GPU makes it faster *and* more accurate.

Thank you. Questions welcome.